

University of Yaoundé I
Department of Physics, Faculty of Science
P.O. Box 812, Yaoundé, Cameroon
myke-vital.sao@facsciences-uy1.cm

August 29, 2026

The Editors
Journal of Cheminformatics
BioMed Central

Dear Editors,

We are pleased to submit our manuscript, **“Pareto-guided Monte Carlo tree search for analysing multi-objective alternatives in antimalarial molecular design”**, for consideration as a Research Article in the *Journal of Cheminformatics*.

Molecular design is a decision problem in which potency, accessibility and risk-related properties can be weighted differently at different stages of lead optimisation. Our manuscript introduces a scaffold-aware Pareto-guided Monte Carlo tree search (MCTS) for antimalarial fragment assembly. The method retains resistance-informed chemotype similarity and a polypharmacology-informed docking score in a candidate archive, allowing alternative profiles to be reviewed rather than reduced to one ranking. These terms are computational design proxies; the study does not present them as biochemical or cellular validation.

Across 20 seeds, the scalar benchmark ranked random search first (0.6724), followed by MCTS (0.6649), the genetic algorithm (0.6453) and greedy search (0.4278). MCTS therefore incurs a measurable scalar-performance cost relative to random search, but the candidate-level analysis retained four non-dominated profiles with hypervolume 1.2366. The largest effects in the separate factorial ablation were associated with the chemistry-informed policy ($\Delta = +0.148$) and Pareto-front maintenance ($\Delta = +0.108$); these are within-ablation effects under a distinct reward configuration.

The manuscript’s central contribution is methodological rather than a claim of scalar algorithmic superiority: it shows how a Pareto archive can preserve chemically distinct decision options while a matched benchmark quantifies the cost of doing so. The public code, molecule sets and analysis outputs are available under the MIT licence at https://github.com/NanaEngo/Malaria_codesV2 for reviewer inspection. We believe this combination of an honest baseline comparison and an explicit decision-oriented objective archive is well aligned with the journal’s computational cheminformatics scope.

This work is original, is not under consideration elsewhere, all authors have approved the submission, and the authors declare no competing financial interests.

We thank you for your consideration.

Myke Vital Sao Temgoua
Corresponding Author

Suggested reviewers:

- Dr. Jan H. Jensen (graph-based genetic algorithms / MCTS for chemical space)
- Prof. Gisbert Schneider (generative molecular design)
- Dr. Ola Engkvist (de novo design, multi-objective optimisation)